

Level 1__ semester II

Robotics

Pre-reading activities :

- 1) Complete the definition of robotics with the right words below :
(embedded system / replicating / behavior / designing / coding)

Robotics is a branch of science which deals with, constructing, and automating mechanical robots that will be capable of humans for a particular task. A robot needs a power source and an that contains high-level which tells the robot to act in a particular

- 2) Listen and complete the missing words.

Pre-Programmed Robots

These are the simplest type of robot. robots work in a fixed situation where they do straightforward, works. For example, a pre-customized robot would be a arm on an assembly line. The arm's responsibility is to that task better than a human.



Humanoid Robots

Humanoid robots will be robots that as well as copy human behavior. These robots perform exercises (like running, bouncing, and carrying objects), and are intended to appear like us, humans, in any event, having human faces and Two of the most well-known of humanoid robots are Hanson Robotics' Sophia and Boston Dynamics' Atlas.

Teleoperated Robots

Teleoperated robots are mechanical whom humans control. These robots work in harsh conditions, climate conditions, and dangerous places. Instances of teleoperated robots are the human-controlled submarines used to fix funnel spills during the BP oil slick or used to recognize landmines in a war zone.



Medical

Clinical and health-care robots incorporate, for example, the da Vinci careful robot and prostheses, and also mechanical

Reading :

Robots: The Next Generation Embedded Computing System

By Lawrence Ricci

As Microsoft's Chairman Bill Gates has boldly predicted, computers will change our lives more in the next 10 years than they have in the last 20. One type of these more influential computers we can expect to see is the intelligent and autonomous sensor platform. In a word: Robots. The development of robots requires no breakthrough in underlying silicon or software technology - the technology is already here. The development of 'smart phone' generation cell phones, has created the CPUs and sensors needed to drive an incredibly diverse range of robotic products. The key is imaginative use of the hardware and software.

Far from being the stuff of science fiction movies, robots are already part of our lives. The Roomba home vacuum cleaner robot from iRobot, for instance, has sold more than one million units. There are robotic sensors monitoring highway traffic flow and providing security in warehouses. Smart sensor platforms called StaBots are measuring global warming by drifting with the movement of European glaciers. High-end robotic toys such as RoboSapiens and Aibo "play" with us to provide entertainment and education. Robotic-assisted surgery with the daVinci® Surgical System allows surgeons to perform complex minimally invasive surgical procedures with precision and accuracy. University programs and robotics clubs are everywhere. Even human-like robots are beginning to appear. Some Japanese companies, concerned about Japan's declining population, have started to develop fully anthropomorphic robots able to work alongside human beings using the same tools and machinery.

One significant use of robot technology is the creation of Unmanned Aerial Vehicles (UAVs). These systems combine sensor technology, flight navigation and control, and communications capability to provide an ability to remotely investigate and monitor large geographic areas. UAVs are being used to monitor forests for fire, rivers for flood, volcanoes for eruption, crops for pests or disease, and snow fields for avalanche.

A major source of UAV development is in the context of military missions. On the battlefield, the value of robots is already inestimable. Convoy vehicles can and will be robotically driven. Manned convoys- or even individual soldiers- can have protective swarms of UAVs overhead, able to detect threats and even intercept them. Robots could guide aircraft or land vehicles on decoy missions to draw out and disclose defenses or ambush.

Applying so much sensor and processor power to a mission may seem a daunting software development task, but it can be accomplished within object-oriented software frameworks already available. Real-time CORBA (common object request broker architecture) implementations exist, simplifying the sharing of data and resources. The Fanuc Handling Pro software is used to create, program and simulate the 3D robot's path in offline mode. The software then sends the information to the robot that can execute the prepared path. Meanwhile some of the world's best minds are trying to solve the toughest problems of robotics, such as visual recognition, navigation and machine learning. And they are succeeding.

Reading comprehension :

1) Decide whether the following statements are true or false. Correct the false ones.

- a. Robotics is a pre-existing technology.
- b. Unmanned vehicles are used solely to supervise war zones.
- c. Anthropomorphic robots took over human jobs in factories.
- d. Creating a suitable software for robotics is a difficult mission.
- e. Visual recognition, navigation and machine learning are some of the unsolved robotics puzzles.

2) Complete the table with the appropriate information from the text.

ROBOT	FUNCTION
StaBots	
Aibo	
daVinci®	

3) Find in the text the hardware and the software used in robots development.

Hardware

Software

4) Match the words with their definitions :

- | | |
|----------------------|--|
| 1. Sensor | a. The science and technology of determining the position of an aircraft with respect to the surface of the Earth. |
| 2. Anthropomorphic | b. A term used to show that the computer or other device is not connected to other devices or network. |
| 3. Flight navigation | c. A device having a microprocessor that estimates the robot's condition and environment. The robots require the extensive information about their environment in order to function effectively. |
| 4. Offline mode | d. A concept showing that a computer program can learn and adapt to new data without human intervention. |
| 5. Machine learning | e. Being capable of reproducing some of the characteristics of man, imitating distinctive features such as appearance and movements. |

Language :

1) Complete each sentence with the suitable collocation (adjective + preposition).

equipped / similar / dangerous / coupled / geared / possible / known / complicated / based

with / for / on / to / toward

1. Robotic has spread like an infection to an extent that so many movies and serials are also its theme.
2. Military robots are a camera to provide soldiers with view of the battlefield and dangerous obstacles.
3. current robotics research is devising robots with a degree of self-sufficiency that will permit mobility and decision-making.
4. Innovation consolidated research and development has catapulted India's scientific position in robotic technology.
5. Robots can also work in conditions that would be humans.
6. South Korea has for many years been..... its cutting-edge intelligent designs, having already deployed robotic teachers, industrial manufacturing workers and service staff.
7. Robots may perform jobs that are too dangerous or too regular workers.
8. Future applications of robotics innovation could also make it South Korea to not only replace a workforce that's rapidly aging and retiring.
9. The challenges facing the robotics industry are those scientists tackled in computing three decades ago.

2) Put the verbs in the right tense or form.

Three 152-centimeter tall robots are nearly ready to begin working as prison guards in the South Korean city of Pohang. The four-wheeled robots **(develop)** as part of the government's \$864,000 robotic guard program. The robots **(design)** to relieve human guards, not replace them, in order to allow guards to focus more on the rehabilitation of inmates. The robots **(equip)** with sensors that allow them to detect behavior among inmates that could lead to violence. Cameras on the robots also **(allow)** guards to communicate remotely with prisoners. The program, scheduled to start in March 2012, **(be)** part of a larger effort to make South Korea the world leader in robotics. Elsewhere, English-speaking robots **(deploy)** as teaching assistants for children in some Korean schools. Many other projects are in the works, such as personal assistant robots **(care)** for the elderly.

Writing:

Your project is to design a robot in the enterprise incubator in your institute. Describe your robot (its type and the purpose of its design)